B. TECH-ME-III-I Sem

23A03506	DYNAMICS LAB	L	T	P	C
		0	1	2	1.5

	Course objectives: The objectives of the course are to
1.	To supplement the principles learnt in kinematics and Dynamics of Machinery.
2.	To understand how certain measuring devices are used for dynamic testing.

Course Outcomes: On successful completion of the course, the student will be able to,	
1.	Ability to demonstrate the principles of kinematics and dynamics of machinery
2.	Determine the Mass moment of inertia, Range sensitivity.
3.	Drawing of Cam profile, determination of torsional, undamped and damped natural frequencies.
4.	Determining of influence of coefficient and balancing of rotating , reciprocating masses.
5.	Verify the laws of springs and forced vibration of cantilever beam.

LIST OF EXPERIMENTS

1. Kinematics of Four Bar, Slider Crank, Crank Rocker, Double crank, Double rocker, Oscillating cylinder Mechanisms.
2. Determination of Mass moment of inertia of Fly wheel and Axle system.
3. Determination of range sensitivity, effort etc., for Watts, Porter, Proell, and Hartnell Governors.
4. Cams – Cam profile drawing, Motion curves and study of jump phenomenon.
5. Determination of torsional natural frequency of single Rotor systems. Un damped and Damped Natural frequencies.
6. Determination of torsional natural frequency of Double Rotor systems. Un damped and Damped Natural frequencies.
7. Multi degree freedom suspension system – Determination of influence coefficient.
8. Determination of torsional natural frequency of single and Double Rotor systems.- Un damped and Damped Natural frequencies.
9. Balancing of rotating masses.
10. Balancing of reciprocating masses.
11. Determination of natural Frequency and verification of Laws of springs
12. Forced Vibration of Cantilever beam – Mode shapes and natural frequencies.